Simulation that models how musical motor skills are retained or lost over a 30-day period based on different practice schedules

```
clear; clc; close all;
%forgetting curve
%stochastic element
%compare three scenarios: the "Weekend Warrior" (one long session),
% the "Consistent Student" (daily short sessions), and the % "Erratic Learner"
(random intervals)

%proficiency percentage over 30days for each schedule. multiline plot
%sensetivity analysis. to show comparision of decay changes for simple and
%complex task
```

```
%parameters
P(1) = 50; % e.g p(1) skill for day 1, p(2)= p(1); copy
R= 10; %reinforcement(boost)
lambda = 0.05;
days = 30;
fatigue = 0.01; % reduces efficiency, if duration > 45
```

```
p = zeros(1, days);
rand = 0.7;
for d=2:days
    p(d) = p(d-1) * exp(-lambda); % forgeting
    if duration > 0
        p(d) = p(d-1) + R; %practice
    end
    %Fatigue
    if duration > 45
        efficiency = max(0, 1- fatigue(duration - 45)); %efficiency can not be zero
    else
        efficiency = 1;
    end


    % Make stochastic, not always perfect some days great practice, some
    % other days distracted
     %randomness, rand(0,1)
   efficiency = efficiency * (0.8 + 0.4*rand);
```

```
end
```

```
Index exceeds the number of array elements. Index must not exceed 1.
```

```
%%weekend warrior
p_weekend = zeros(1, days);
p_consistent = zeros(1, days);
p_erratic = zeros(1, days);
% % initialize
p_weekend(1) = 50;
p_consistent(1) = 50;
p_erratic(1) = 50;

%% weekend warrior
for d = 2:days
p_weekend(d) = p_weekend(d-1) *exp(-lambda);
   if mod(d,7) ==6
       duration = 120;
   else
       duration =0;
   end

 if duration > 45
    efficiency = max(0, 1- fatigue*(duration - 45));%efficiency can not be zero
 else
     efficiency = 1;
  end

    efficiency = efficiency * (0.8 + 0.4*rand);

    if duration > 0
        p_weekend(d) = p_weekend(d-1) +R *efficiency;
    end







end
```

Consistent Student

```
p_consistent(d) = p_consistent(d-1)*exp(-lambda);
duration = 30;
```

```
if duration > 45
    efficiency = max(0, 1- fatigue*(duration - 45));%efficiency can not be zero
 else
     efficiency = 1;
end

    efficiency = efficiency * (0.8 + 0.4*rand);

    if duration > 0
        p_consistent(d) = p_consistent(d-1) +R *efficiency;
    end
```

Erratic Learner

```
p_erratic(d) = p_erratic(d-1)*exp(-lambda);

if rand > 0.5 % 50% true, 50% False
    duration = randi([20,90]);

else
    duration =0;
end

if duration > 45
    efficiency = max(0, 1- fatigue*(duration - 45));%efficiency can not be zero
 else
     efficiency = 1;
end

    efficiency = efficiency * (0.8 + 0.4*rand);

    if duration > 0
        p_erratic(d) = p_erratic(d-1) +R *efficiency;
    end
```

Explaining models

Forgetting: Skill decreasing | exponential decay

Reinforcement: practice increases skill

Fatigue: Long sessions not effective

Randomness: Human inconsistency

```
%Visualization
```

```
%Sensetivity Analysis
```